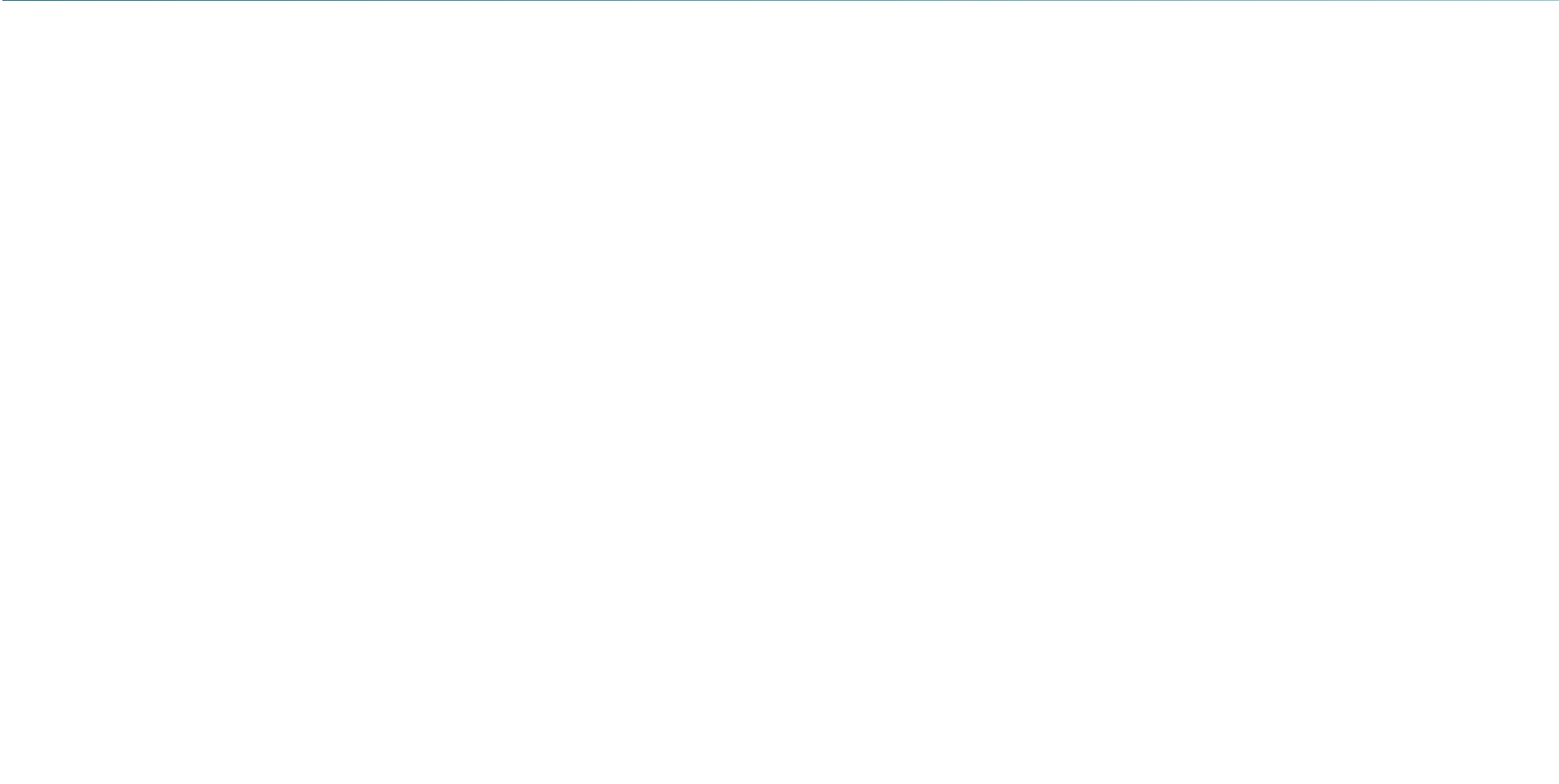


AhamX Bodhi

Where I Becomes Infinite

What is Generative AI?

Module 1: Practical GenAI Foundations
Prof. Sashikumaar Ganesan





Session Overview

Session Details

- **Domain:** Generative AI – Theory
- **Duration:** 10 minutes
- **Bloom's Level:** B2 (Understand)
- **Difficulty:** L2 (Intermediate)
- **Proficiency Target:** Early Competent

Foundational Entry Point

This is the **first concept** in the GenAI curriculum. No prior knowledge of generative AI is assumed. This session establishes the foundation for everything that follows.



Learning Objectives

By the End of This Session, You Will Be Able To

1. **Define** Generative AI and distinguish it from traditional AI approaches
2. **Explain** how generative models learn to produce new content
3. **Identify** the major categories of generative AI outputs
4. **Recognize** the key architectures powering modern GenAI systems
5. **Describe** why Generative AI represents a paradigm shift in computing



Prerequisites & What You Will Gain

Prerequisites

- None – this is a foundational entry point
- Basic familiarity with computers is helpful but not required
- Curiosity about AI is all you need

What You Will Gain

- A clear mental model of what GenAI is and is not
- Vocabulary to discuss GenAI confidently
- Context for every subsequent concept in this program

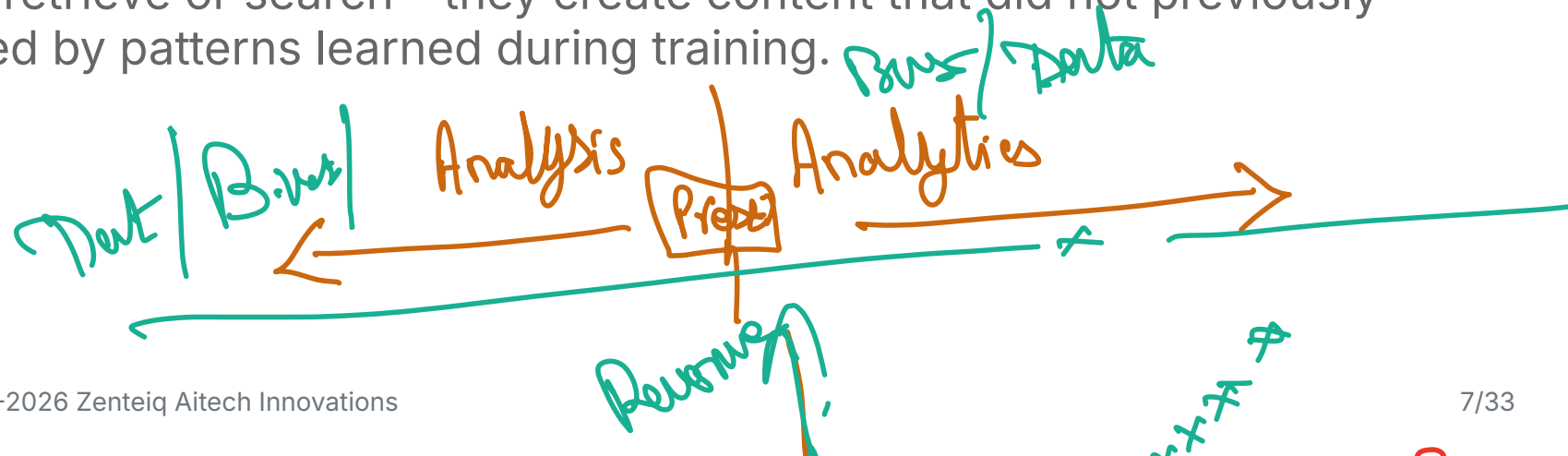


What is Generative AI?

Definition

Generative AI refers to artificial intelligence systems that can create new content – including text, images, audio, video, and code – by learning patterns and structures from existing data. Rather than simply analyzing or classifying inputs, generative models produce novel outputs that resemble the training data.

The word *generative* is key: these systems generate something new. They do not just retrieve or search – they create content that did not previously exist, guided by patterns learned during training.



Traditional AI vs Generative AI

Traditional AI (Analytical)

- **Classifies** data into categories
- **Predicts** numerical values
- **Detects** anomalies or patterns
- Answers: "What is this?"
- Example: spam filter, fraud detection

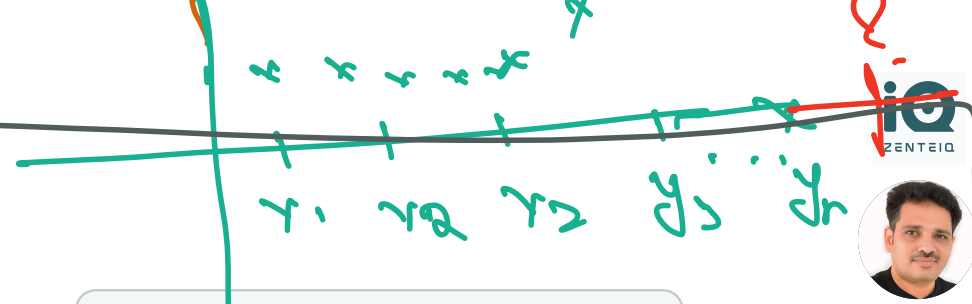
Tools

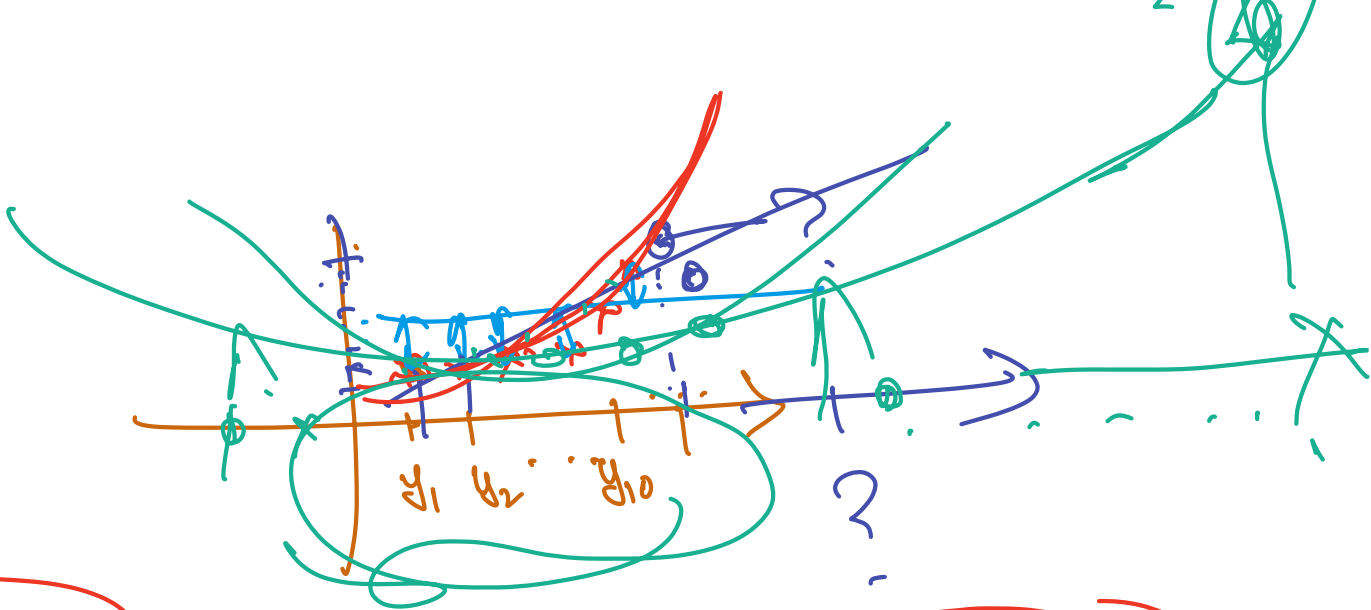
Generative AI (Creative)

- **Creates** new text, images, code
- **Generates** realistic content
- **Transforms** content across formats
- Answers: "What could this be?"
- Example: ChatGPT, DALL-E, Copilot

Physics Exam

A generic AI model





y	R
y_1	r_1
y_2	r_2
$\vdots$	$\vdots$
y_{10}	r_{10}
$\vdots$	$\vdots$
Q	$?$

$S \subset R$

$$r(y) = \underline{\underline{r}}$$

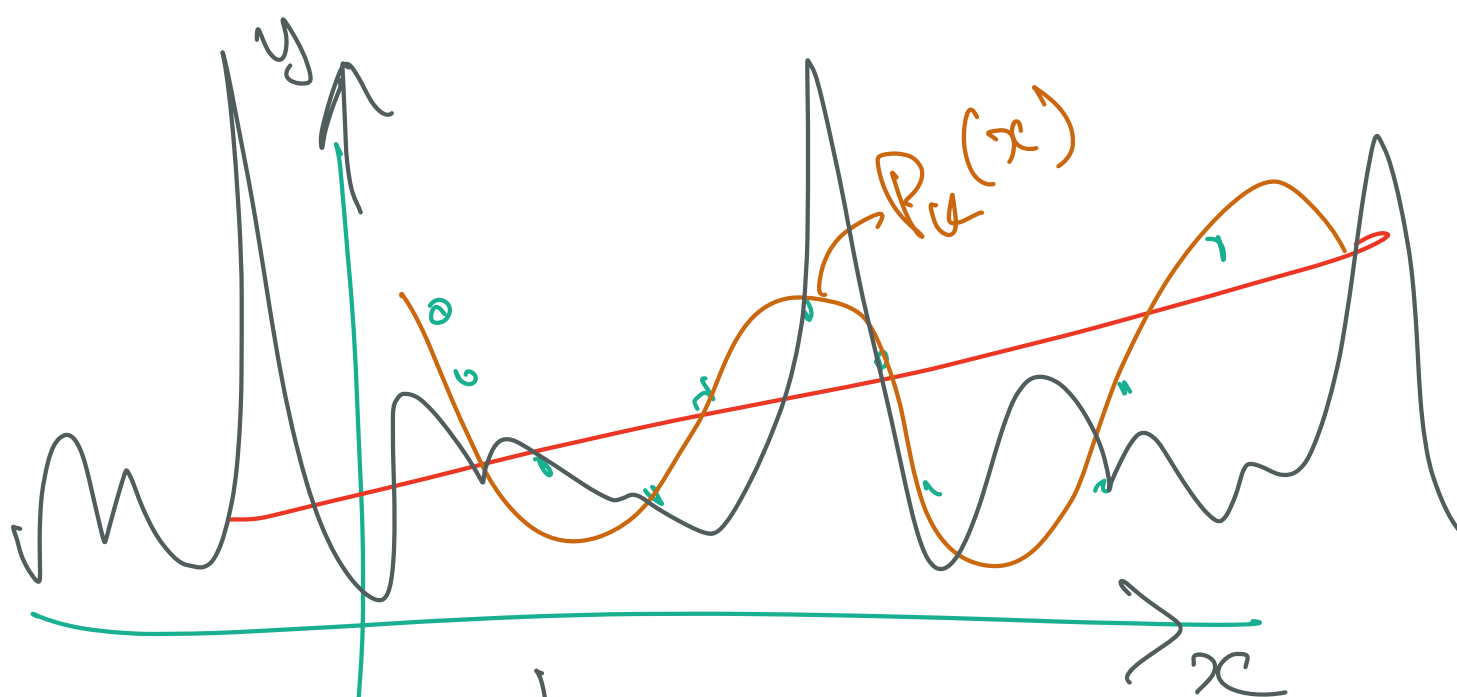
$$r(y) = w_1 y + b$$

$$r(y) = w_1 y + w_2 y^2 + b$$

Polynom Degree: $r_k(y) = w_1 y^k + \dots + b$

$$y(x) = w_1 x + b$$

$x=0$





How Does Generative AI Work?

The Core Mechanism

1. **Training:** The model ingests vast amounts of data (text, images, code) and learns the statistical patterns, structures, and relationships within that data
2. **Learning Representations:** The model builds internal representations that capture the "rules" of the data – grammar, visual features, logical structures
3. **Generation:** Given a prompt or input, the model uses its learned representations to produce new content that is consistent with the patterns it learned

Think of it like learning a language by reading millions of books – you internalize the patterns and can then write new sentences you have never seen before.



What Can Generative AI Create?

Text & Language

- Articles, essays, and reports
- Code in any programming language
- Translations across languages
- Summaries and paraphrases

Visual & Media

- Photorealistic images from text
- Video clips and animations
- Music and audio compositions
- 3D models and designs



What Can Generative AI Create? (continued)

Structured Data

- Synthetic datasets for training
- Database queries from natural language
- Structured reports and tables
- API schemas and configurations

Scientific & Technical

- Molecular structures for drug discovery
- Protein folding predictions
- Mathematical proofs and derivations
- Engineering design alternatives



Key Architectures Behind Generative AI

The Major Model Families

- **Transformers** – The dominant architecture for language models (GPT, BERT, T5). Use self-attention to process sequences and generate text token by token
- **Diffusion Models** – Power modern image generation (Stable Diffusion, DALL-E 3). Gradually denoise random noise into coherent images
- **GANs (Generative Adversarial Networks)** – Two networks compete: a generator creates content, a discriminator evaluates it. Pioneered realistic image synthesis
- **VAEs (Variational Autoencoders)** – Learn compressed representations of data and generate new samples from that latent space



The Transformer Revolution

Why Transformers Changed Everything

- Introduced in the 2017 paper "Attention Is All You Need"
- **Self-attention mechanism** allows the model to consider all parts of the input simultaneously, not just sequentially
- Scales efficiently to billions of parameters
- Powers virtually all modern LLMs: GPT-4, Claude, Gemini, Llama

Transformers made it possible to train models on internet-scale data and produce remarkably coherent, contextual outputs. This single architecture underlies most of what we call "Generative AI" today.



Why This is a Paradigm Shift

A Fundamental Change in Computing

- **Before GenAI:** Software followed explicit rules written by programmers. Outputs were deterministic and predictable
- **With GenAI:** Models learn implicit rules from data. Outputs are probabilistic and can be creative, surprising, or even incorrect
- **The shift:** From programming *instructions* to providing *examples*

This shift means that anyone who can describe what they want in natural language can now “program” an AI system – democratizing access to computational creativity.





GenAI Application: Large Language Models

LLMs in Production Today

- **OpenAI GPT-4 / GPT-4o:** Powers ChatGPT, Microsoft Copilot, and thousands of enterprise applications via API
- **Anthropic Claude:** Used in safety-critical applications with constitutional AI alignment and long-context reasoning
- **Google Gemini:** Integrated across Google products – Search, Workspace, Android – and available as API for developers
- **Meta Llama 3:** Open-weight models enabling on-premises deployment and custom fine-tuning without API dependency

These models process billions of tokens daily, serving millions of users across every industry from healthcare to finance.



GenAI Application: Image Generation

Visual Content Creation at Scale

- **Midjourney:** Produces photorealistic and artistic images from text descriptions; widely used by designers and marketing teams
- **DALL-E 3 (OpenAI):** Tightly integrated with ChatGPT for conversational image creation; excels at following complex prompts
- **Stable Diffusion (Stability AI):** Open-source diffusion model enabling local deployment, custom training, and community extensions
- **Adobe Firefly:** Commercially safe image generation trained on licensed content; integrated into Photoshop and Creative Cloud

Image generation has moved from research curiosity to production tool in under three years.



GenAI Application: Code Generation

AI-Powered Software Development

- **GitHub Copilot:** Autocompletes code in real-time inside VS Code and JetBrains IDEs; trained on public code repositories
- **Anthropic Claude Code:** Command-line agent for agentic coding – delegates entire coding tasks directly from the terminal
- **Cursor / Windsurf:** AI-native code editors that integrate LLMs deeply into the development workflow
- **Amazon CodeWhisperer:** Enterprise code generation optimized for AWS services and security scanning

Studies show AI-assisted developers complete tasks 30–55% faster, fundamentally changing how software is built.



GenAI Application: Multimodal AI

Beyond Single-Modality Models

- **GPT-4 Vision:** Accepts images and text together; can describe photos, read charts, solve visual math problems
- **Google Gemini:** Natively multimodal – processes text, images, audio, and video in a single model
- **OpenAI Sora:** Generates realistic video from text prompts; demonstrates understanding of physical world dynamics
- **ElevenLabs:** Produces human-quality speech synthesis and voice cloning from text input

Multimodal models are converging toward systems that understand and generate across all human communication modalities.





GenAI Application: Enterprise & RAG Systems

Grounding LLMs in Enterprise Knowledge

- **Retrieval-Augmented Generation (RAG):** Combines LLMs with vector databases to answer questions using company-specific documents
- **Pinecone / Weaviate / Qdrant:** Vector databases that store embeddings for fast semantic search, enabling LLMs to retrieve relevant context
- **LangChain / LlamaIndex:** Frameworks that orchestrate the retrieval-generation pipeline for production RAG applications
- **Enterprise Copilots:** Custom AI assistants trained on internal knowledge bases (Salesforce Einstein, ServiceNow, Slack AI)

RAG systems solve the “hallucination problem” by grounding LLM responses in verified, up-to-date enterprise data.



GenAI Application: Agentic AI Systems

From Chatbots to Autonomous Agents

- **AI Agents** combine LLMs with tools, memory, and planning to perform multi-step tasks autonomously
- **ReAct Framework:** Agents reason about what to do (Thought), take action (Act), and observe results (Observation) in a loop
- **Tool Use:** Agents call APIs, search the web, execute code, query databases – extending LLM capabilities far beyond text generation
- **Production Systems:** OpenAI Assistants API, Anthropic tool use, LangGraph, and CrewAI power real-world agent deployments

Agentic AI represents the next frontier: AI that does not just answer questions but **completes tasks** end-to-end.





Skills Developed in This Session

Conceptual Skills

- GenAI conceptual literacy
- Architecture awareness (Transformers, Diffusion, GANs)
- Paradigm shift understanding

Applied Skills

- GenAI output type identification
- Model family classification
- Industry application mapping



Professional Relevance

Who Benefits from This Knowledge

- **ML Engineers:** Foundation for understanding models they will build, fine-tune, and deploy
- **AI Engineers:** Context for integrating LLMs and generative models into production applications
- **Data Scientists:** Framework for evaluating when to apply generative vs discriminative approaches
- **Product Managers:** Vocabulary and mental models for making informed decisions about GenAI products
- **Software Engineers:** Understanding of the tools (Copilot, Claude Code) transforming their daily work



Knowledge Graph: Prerequisites

No Prerequisites Required

- This is the **foundational entry point** for the entire GenAI curriculum
- No prior knowledge of machine learning, deep learning, or AI is assumed
- The course overview session provides helpful context but is not required

This concept serves as the “starting node” in the knowledge graph. Everything else in the program builds from here.



Knowledge Graph: What This Unlocks

Directly Enabled Concepts (Part 1)

- **Generative vs Discriminative Models** – Deepens the distinction introduced here
- **Organizational AI Strategy** – Applies GenAI understanding to business contexts
- **GenAI Use Cases & Applications** – Explores specific industry applications in detail
- **Risks & Limitations of GenAI** – Examines what can go wrong and how to mitigate it
- **Five-Level Framework for GenAI** – Provides a maturity model for GenAI adoption



Knowledge Graph: What This Unlocks (continued)

Directly Enabled Concepts (Part 2)

- **What is a Language Model?** – Dives into the specific model type powering LLMs
- **What is Prompt Engineering?** – Introduces how to communicate effectively with GenAI
- **How LLMs Learn from Data** – Explores the training process behind generative models
- **Setting Up Development Environment** – Prepares you to start building with GenAI tools



Related Concepts & Connections

Within This Program

- Transformer Architecture
- Tokenization Fundamentals
- Attention Mechanisms
- RAG Systems
- Agentic AI

Broader AI Landscape

- Deep Learning fundamentals
- Neural network architectures
- Natural language processing
- Computer vision
- Reinforcement learning





Key Takeaways

Remember These Five Points

- Generative AI **creates new content** rather than just analyzing or classifying existing data
- The **Transformer architecture** is the dominant engine behind modern LLMs like GPT-4 and Claude
- GenAI outputs span **text, images, code, audio, video**, and structured data
- This technology represents a **paradigm shift** from rule-based programming to learning from examples
- GenAI is already in **production** across every major technology company and industry

Next Steps



Continue Learning

- Next session: Generative vs Discriminative Models
- Explore one GenAI tool hands-on (ChatGPT, Claude, or Gemini)
- Read the original "Attention Is All You Need" abstract

Proficiency Check

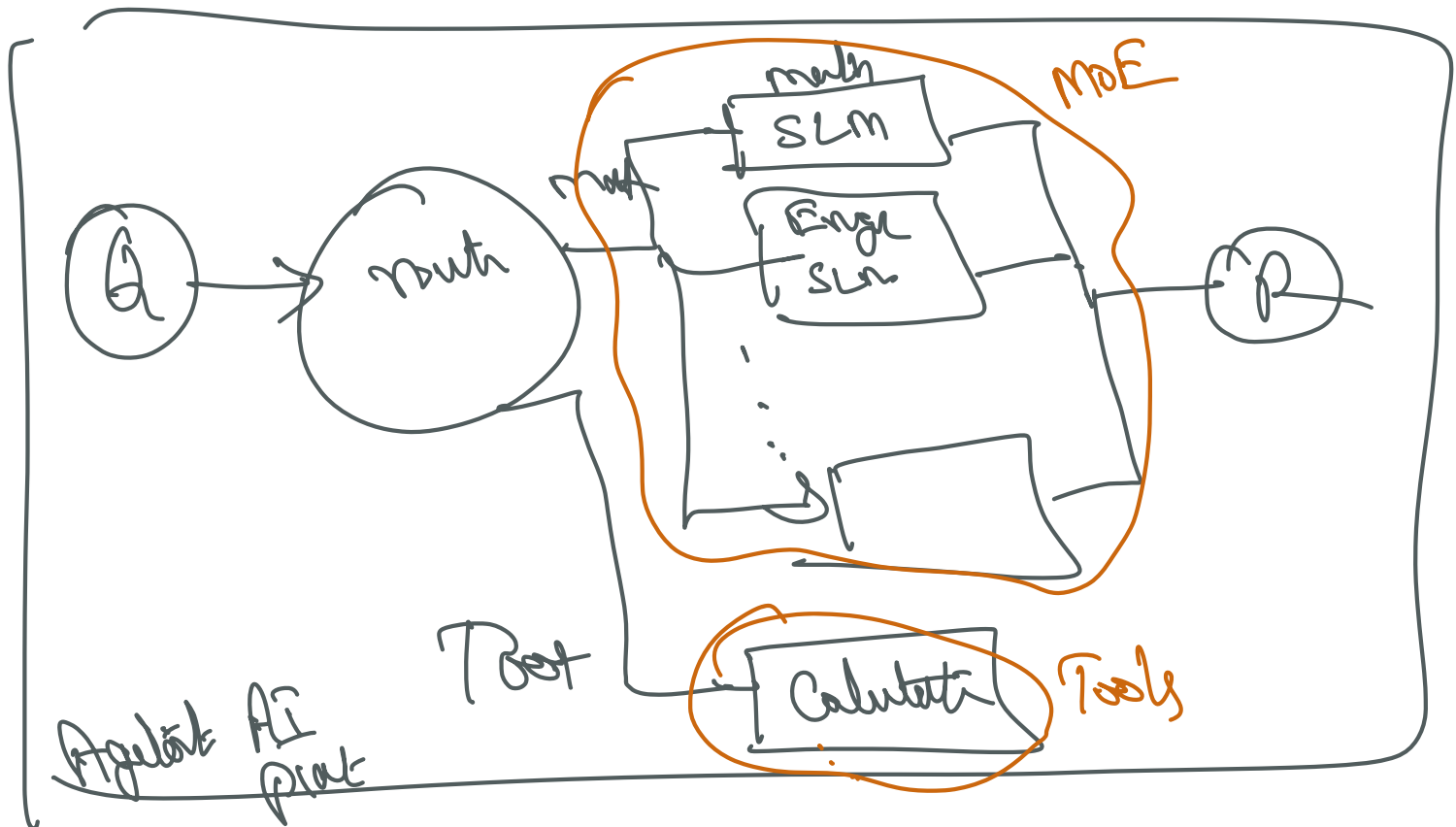
- Can you define GenAI in one sentence?
- Can you name three types of content GenAI can produce?
- Can you explain why Transformers matter?

What is Generative AI?

*You now have the foundation.
Next: Generative vs Discriminative Models*

Trad ML $\leftarrow$ Tabular Data

Gen AI $\leftarrow$ language [Code]



T_1 T_2 T_3 $\leftarrow \text{Given } (T_1, T_2) \rightarrow T_3?$
 I like

I like Playing T_4

$\text{Given } (T_1, T_2, T_3) \rightarrow T_4?$

